INTERNATIONAL MONETARY FUND

# REPUBLIC OF TAJIKISTAN

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## STAFF REPORT FOR THE 2015 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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*Based on the external low income country (LIC) debt sustainability analysis (DSA), Tajikistan's risk of debt distress is assessed to be low, upgraded from its previous moderate rating.[1] Assessing debt sustainability with thresholds used for medium policy performers—due to an improvement in Tajikistan's Country Policy and Institutional Assessment (CPIA) rating—and using the higher unified discount rate of 5 percent in place for LIC DSAs since October 2013, while also accounting for a lower initial debt position since Tajikistan has borrowed less in recent years than projected in the last DSA, moves Tajikistan from moderate to low risk of debt distress.[2] Despite an increase in borrowing capacity implicitly expressed by increases in the thresholds and the lower initial debt position, Tajikistan's external position is projected to remain sensitive to a number of shocks, especially given the deteriorating macroeconomic environment, limited policy action to tackle existing vulnerabilities, and very thin external and fiscal buffers. Stress tests within the public DSA demonstrate the sensitivity of the fiscal position with respect to a slightly lower long run growth rate.*

[1] The DSA has been produced jointly by World Bank and IMF staff, in consultation with Asian Development Bank staff. The fiscal year for Tajikistan is January 1–December 31.

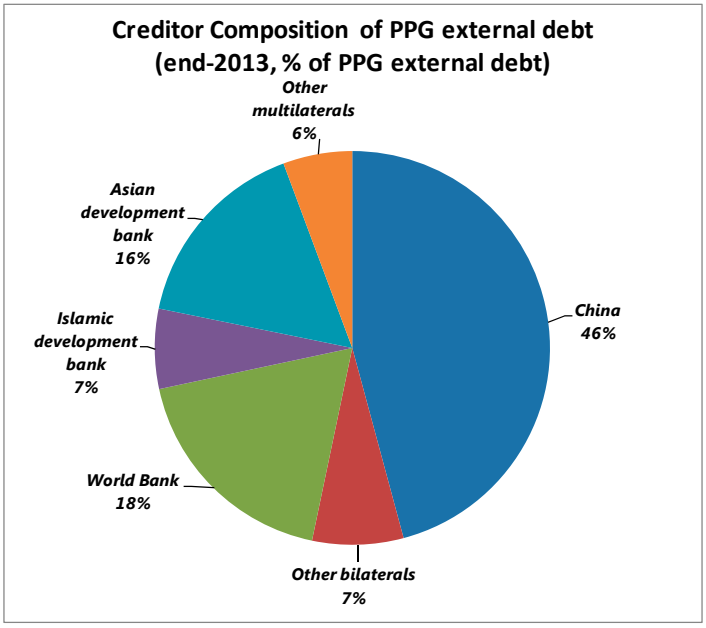
[2] On October 11, 2013, the Executive Boards of the IMF and World Bank raised to 5 percent (from 3 percent) the discount rate used to calculate the present value of the external debt of low-income countries using the LIC Debt Sustainability Framework. This change came into effect after the previous (July 19, 2013) LIC DSA for Tajikistan and has the effect of lowering the present value of external debt in the current DSA compared to the previous DSA, all other things equal.

# BACKGROUND

**1. Tajikistan has remained vulnerable to external and growth shocks though its public debt stock declined from about 65 percent of GDP in 2003 to 29 percent of GDP in 2013.** Most of the public debt is denominated in foreign currency—25.3 percent of GDP in 2013.

**2. Public and publicly guaranteed (PPG) external debt at end 2013 dropped to 25.3 percent of GDP, a reduction from the previous year's level by 3 percent of GDP (Table 1).** This was mainly due to a combination of continued strong real GDP growth and lower external financing. Tajikistan's PPG external debt is largely concessional and denominated in foreign currency, and held in broadly equal parts between multilateral and bilateral creditors, with the share of China quite prominent at around 46 percent (text chart).[3] Private external debt was 18.3 percent of GDP in 2013 and largely long-term and denominated in foreign currency.

**3. Domestic debt continues to constitute a limited part of overall public debt owing to low domestic financing needs as the primary balance, excluding the externally financed public investment program, has usually been in surplus.** Domestic debt stood at close to 4 percent of GDP (13 percent of total public debt) at end-2013, consisting mostly of non-tradable government securities held by the National Bank of Tajikistan (NBT). The government securities market remains shallow. Since 2009, the government has undertaken issuance of treasury-bills, but only in small amounts, with short maturity, and at rates below the NBT refinancing rate.[4]




[3] Information is provided by the country authorities.

[4] In 2013, the net total issuance (excluding issuances for the Agroinvestbank resolution and NBT recapitalization) was - 0.02 percent of GDP based on auction amounts.

| Tajikistan: External and Overall Public Debt, 2013–16 | | | | |
|---|---|---|---|---|
| | 2013 Act. | 2014 Est. | 2015 Proj. | 2016 Proj. |
| | (Percent of GDP, unless indicated otherwise) | | | |
| **Total External Debt** | 43.6 | 42.5 | 46.7 | 47.5 |
| (Billion of US dollars) | *3.7* | *3.8* | *3.9* | *4.3* |
| Public and Publically Guaranteed (PPG) Debt | 25.3 | 24.5 | 25.7 | 26.5 |
| (Billion of US dollars) | *2.1* | *2.1* | *2.1* | *2.4* |
| Private Debt | 18.3 | 18.0 | 21.0 | 21.0 |
| (Billion of US dollars) | *1.6* | *1.7* | *1.8* | *1.9* |
| **Total Domestic Public Debt** | 3.9 | 3.7 | 3.4 | 3.2 |
| (Billion of US dollars) | *0.3* | *0.3* | *0.3* | *0.3* |

**4. The current DSA—which is based on thresholds for medium policy performers and the higher (5 percent) discount rate applicable to LIC DSAs since October 2013, and also accounts for the lower initial debt position—concludes that Tajikistan's risk of debt distress is reclassified to low from moderate (see Figure 1).**[5] [6] Due to an improvement in its three-year average CPIA rating through 2013, Tajikistan's policies and institutions are now classified as those corresponding to a "medium performer". Institutional changes do not, however, appear to be deep and may be difficult to sustain over time given that structural reforms continue to lag. Nonetheless, contingent on the improved three-year average CPIA and the higher discount rate of 5 percent applied to external debt in LIC DSAs since October 2013, in the current assessment all external PPG debt burden indicators are projected to remain below their respective thresholds under the baseline and stress scenarios, indicating Tajikistan's low risk of debt distress.

# MACROECONOMIC ASSUMPTIONS

**5. The macroeconomic framework over the medium term presents a deteriorated picture compared with the one shown in the previous DSA** (Text table and Box 1). The current macro framework assumes a slowdown in growth, a higher initial GDP deflator, and a larger current account deficit over the short-term. A sharp fall in exports of aluminum and cotton in 2014, combined with an accelerating decline in remittance inflows through 2014–15 (remittances have been equivalent to

[5] The Joint Bank-Fund Staff Guidance Note on Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries (November 2013) defines that remittances must be treated in the base case if remittances are both greater than 10 percent of GDP and greater than 20 percent of exports of goods and services. Both ratios should be measured on a backward-looking, three-year average basis. The remittance-adjusted thresholds for **medium policy performers** like Tajikistan are revised up to 120 (previously 80) and 36 (previously 27) for the PV of debt to exports and GDP, respectively. For debt service, the remittance-adjusted threshold is raised to 16 percent of exports (previously around 12 percent). Thresholds for the PV of debt-to-revenues and debt service–to-revenues are now 250 and 20 percent (previously 200 and 18), respectively.

[6] The latest three-year average of the World Bank's CPIA rating (2011–13) for Tajikistan is 3.34, with the CPIA rating of 2013 at 3.31. According to the Joint Bank-Fund Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework (November 2013), Tajikistan's policies and institutions are now assessed as those corresponding to a "medium performer" due to being above the threshold of 3.25 over two consecutive years.

40-50 percent of GDP in recent years and originate almost entirely from Russia), result in a larger current account deficit (-9.1 and -7.1 percent in 2014 and 2015, respectively) over the short-term even after allowing for the projected import compression from exchange rate adjustment and lower growth.[7] The U.S. dollar deflator is projected to decrease in 2015 in view of substantial exchange rate depreciation.

**6. The PPG debt-to-GDP ratio is estimated to be slightly lower than the one in the previous DSA, mostly due to lower public investment project (PIP) disbursements over the short term.** Over the medium term, this debt ratio is expected to reach the same level as in the previous DSA based on an expected upward trend of PIP disbursements from bilateral and multilateral creditors and lower growth projections. This is reflected in higher fiscal expenditure as a share of GDP through the medium term, leading to a larger (smaller) fiscal deficit (surplus), including (excluding) PIP spending. Tax revenues (though higher as a share of GDP in this DSA based on revenue collections to date) are not expected to offset this effect due to the large balance of payments and growth shock. The present value (PV) of PPG external debt relative to GDP, exports, and revenues are lower than those in the previous DSA due to the upward change in the discount rate to 5 percent. The increase in the discount rate also increases the grant element of new borrowing, meaning that Tajikistan is borrowing on relatively more favorable terms from external creditors.

**Tajikistan: Baseline External DSA Compared to Previous DSA, 2014–16**

| | Previous DSA[1] | | | Current DSA | | |
|---|---|---|---|---|---|---|
| | 2014 Proj. | 2015 Proj. | 2016 Proj. | 2014 Proj. | 2015 Proj. | 2016 Proj. |
| Stock of public and publicly-guaranteed (PPG) external debt | | | | | | |
| Percent of GDP | 27.0 | 27.7 | 27.4 | 24.5 | 25.7 | 26.5 |
| Present value (PV) of PPG external debt to GDP (percent) | 15.2 | 14.9 | 15.2 | 14.7 | 15.4 | 14.7 |
| PV of PPG external debt to revenues (percent) | 63.9 | 61.9 | 63.8 | 54.7 | 63.1 | 60.6 |
| PV of PPG external debt to exports (percent) | 96.1 | 94.0 | 98.2 | 95.4 | 80.5 | 75.5 |
| PPG external debt service to revenues (percent) | 5.0 | 4.8 | 5.0 | 4.7 | 6.0 | 5.7 |
| PPG external debt service to exports (percent) | 7.6 | 7.4 | 7.7 | 8.1 | 7.6 | 7.1 |
| Discount rate (percent) | 3.0 | 3.0 | 3.0 | 5.0 | 5.0 | 5.0 |
| | (Percent of GDP, unless indicated otherwise) | | | | | |
| Real GDP (percentage change) | 5.8 | 5.7 | 5.7 | 6.7 | 3.0 | 4.1 |
| GDP Deflator (cumulative, percentage change) | 7.0 | 7.0 | 7.0 | 5.5 | 12.7 | 8.0 |
| **Fiscal** | | | | | | |
| Revenue and grants | 24.9 | 24.6 | 25.0 | 28.4 | 26.2 | 26.5 |
| Tax revenue | 19.9 | 19.9 | 20.1 | 22.8 | 20.5 | 20.6 |
| Expenditure and net lending | 26.2 | 25.6 | 26.5 | 28.3 | 28.0 | 28.7 |
| Overall balance (excl. PIP and stat. discrepancy) | 0.4 | 0.4 | 0.4 | 1.0 | -0.3 | -0.1 |
| Overall balance (incl. PIP and stat. discrepancy) | -1.3 | -1.0 | -1.4 | 0.1 | -1.8 | -2.2 |
| Grant element of new external borrowing (percent)[2 3] | 17.0 | 17.0 | 17.0 | 33.5 | 39.5 | 40.9 |
| **Balance of payments** | | | | | | |
| Exports of goods and services (in million US$) | 1,540 | 1,650 | 1,761 | 1,423 | 1,630 | 1,791 |
| Imports of goods and services (in million US$) | 5,918 | 6,326 | 6,855 | 5,747 | 4,414 | 4,469 |
| Remittance inflows (percentage change) | 8.9 | 6.8 | 9.0 | -8.3 | -35.0 | 0.0 |
| Current account balance | | | | | | |
| Millions of U.S. dollars | -207 | -236 | -257 | -839 | -604 | -533 |
| Percent of GDP | -2.2 | -2.4 | -2.4 | -9.1 | -7.1 | -5.8 |

Source: authorities, IMF, and World Bank staff.

[1] July 2013. PV of PPG debt was computed based on a discount rate of 3 percent.

[2] Includes publicly-guaranteed external borrowing.

[3] Grant element in previous DSA was computed by using a discount rate of 3 percent.

[7] The change in the current account deficit also reflects an update in the base year: the current account balance in 2013 was revised to -2.9 percent of GDP from -1.4 percent of GDP after adjusting for unrecorded oil imports in 2012 and 2013.

## Box 1. Tajikistan: Macroeconomic Assumptions

**Real GDP growth** is projected at 3 and 4 percent in 2015 and 2016, respectively, and at 5 percent over the medium and long term, below the ten-year historical average of 7¼ percent (2004–13) and similar to medium-term projections in the previous DSA (around 5¾ percent). Adverse projected global conditions, particularly the slowdown in the Russian economy together with the large ruble depreciation, are expected to weigh on growth over the near term through lower remittance inflows and slowing private consumption. The outlook incorporates stable fiscal balances (excluding externally financed projects) over the medium term, a moderate pace of construction of the Rogun hydro power plant (HPP; to be reassessed following the authorities' completion of a financing plan), exchange rate depreciation, and little progress on structural reforms.

**The U.S. dollar GDP deflator** is projected to increase by 1¾ percent in 2014. It then decreases by 10½ percent in 2015, largely reflecting substantial exchange rate depreciation. The annual change in this deflator is expected to return to about 3½ percent through 2034.

**The current account deficit** is estimated to have increased to 9 percent of GDP in 2014 due to a reduction in remittances—tied particularly to spillovers from Russia—and a fall in cotton and aluminum exports. It is projected to improve to 3½ percent of GDP over the medium term, driven by a return to growth in remittances, recovery of exports, including due to FDI projects, and moderate import growth consistent with the assumed pace of Rogun construction and economic growth.

**Migrants' remittances,** which were equivalent to 50 percent of GDP in 2013 (gross), fell by 8.7 percent in U.S. dollar terms in 2014. Remittance flows in US dollars are expected to deteriorate further to -35 percent in 2015, assuming a contraction in Russia and the impact of large ruble depreciation, and recover to 7 percent growth over the medium and to 9 percent growth over the long term (well below recent double-digit averages).

**Reserve coverage** is initially projected to decline in 2015 to 1.2 months of import cover from 1.4 in 2014, and then build up to about 2.0 months of imports by 2019, and increase gradually to about 3¼ months of imports over the long run.

**The fiscal deficit** (overall primary deficit) is expected to grow over the near and medium term, driven by externally financed public investment projects (PIP). However, excluding externally financed PIP, the fiscal balance improved in 2014 relative to the 2013 outcome and is expected to be near balance, consistent with medium-term debt sustainability and with the goal of protecting fiscal buffers, except in the adjustment period 2015-16. The fiscal path also assumes moderate domestic capital spending on the Rogun project, at similar levels in percent of GDP as in the past several years. **Total interest expenditures** are expected to stabilize at around 0.5 percent of GDP over the medium term, while amortization is expected to decline gradually from 1.1 percent in 2017 to 0.6 percent over the long term.

**Financing.** External and domestic PPG debt is projected to account for about 89 (91) percent and 11 (9) percent, respectively, of total public debt over the medium (long term). The share of domestic PPG debt over the projection period is expected to be higher than the ten-year historical average of 6 percent of total public debt (2004-13) in view of the need to raise domestic financing to help service external debt. External PPG is expected to be largely concessional and denominated in foreign currency, in line with its trend over the last 10 years. Following the government's framework, annual expenditure of about 2 percent of GDP during 2016–19 to finance the construction of the Rogun HPP is incorporated, with the assumed externally financed component treated as unidentified financing in the framework.[8]

[8] The government's current medium-term fiscal framework, and staff projections, include annual expenditure of about 2 percent of GDP ($220 million for 2015) to complete basic infrastructure at the Rogun HPP site; this infrastructure is required to divert the Vaksh river to allow for eventual dam construction. While the government has indicated that it would like to build the full-height Rogun dam (remaining estimated cost of $5-6 billion), no sources

(continued)

# EXTERNAL DSA

**7. Under the baseline scenario in which thresholds for medium policy performers are applied, all external debt burden indicators remain below the respective thresholds (see Figure 1 and Table 1a).** External debt stock ratios are expected to be on a gradual upward trajectory over the medium term, and then steady over the long term. Furthermore, external debt service ratios are expected to be on a gradual downward trajectory over the medium term and then steady over the long term. During the projection period, debt service payments continue to be manageable, albeit with a gradual increase during the years when principal payments on loans from China continue to fall due (starting from 2012).[9]

**8. Unlike, the previous DSA—which demonstrated protracted breaches in all external debt stock indicators in the standard stress tests—there are no threshold breaches under the stress scenario in the current DSA (Figure 1 and Table 1b).** Under the baseline scenario, the PV of debt-to-GDP + remittances, the PV of debt-to-exports + remittances, and the PV of debt-to-revenue are now projected to be on a gradual upward path while remaining below their respective thresholds**.** Moreover, the PV of debt-to-GDP + remittances and the PV of debt-to-exports + remittances are projected, especially under the most extreme shock scenario (an one-time drop in export value growth in 2015-16), to remain high after a sharp increase, indicating some sensitivity of external debt indicators, while staying below the new threshold for medium policy performers. In the previous DSA, the PV of debt-to-GDP + remittances was projected to breach the indicative thresholds for weak policy performers substantially for a protracted period of time and the PV of debt-to-exports + remittances and PV of debt-to-revenue were forecasted to breach significantly for a short period of time. Nonetheless, Tajikistan's external position continues to be sensitive to interest rate shocks, remittance-led growth deviations, and large exchange rate depreciation. This suggests the need to broaden the revenue and export bases through diversification and structural reforms (e.g. implementation of a dividends policy for SOEs, bank restructuring, measures to strengthen the business climate) to ensure greater competitiveness and to limit the potential for contingent risks to materialize. In view of the above, and notwithstanding the technical upgrade in the debt distress rating, the authorities should remain cautious in contracting and guaranteeing new debt, particularly on non-concessional terms, and should conduct and assess carefully cost-benefit studies of large-scale infrastructure projects. The authorities should also develop the domestic debt market as an alternative source of market-based financing and to facilitate the development of a robust monetary policy framework.

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of financing or other cash flows have been identified to complete the project. The full cost of Rogun is, therefore, not included in the baseline macroeconomic framework or the DSA.

[9] Information is provided by the country authorities. The identified debt repayment obligations to China (one of the biggest creditors) and other multilateral and bilateral creditors are covered by sufficient foreign exchange inflows in the capital accounts.

# PUBLIC DSA

**9. Public debt burden indicators (including domestic public debt) such as the PV of debt-to-GDP ratio and the debt-to-revenue ratio are also expected to be steady over the medium term and on a gradual upward trajectory over the long run under the baseline scenario (see Figure 2 and Table 2a).** Contrary to the previous DSA, the PV of the public debt-to-GDP ratio would remain stable over the medium term and increase gradually over the long term, mostly due to lower growth projections underlying the current DSA. In a similar vein, the PV of debt-to-revenue is projected to be on an upward trend over the long term.

**10. Standard stress tests indicate the existence of vulnerabilities in public debt (Figure 2).** The debt burden indicators are sensitive to permanently lower GDP growth, indicating the need to increase fiscal buffers. Also, the indicators deviate noticeably from the baseline under the scenario of higher other debt creating flows (10 percent of GDP increase in other debt creating flows in 2015 which more than captures plausible shocks arising from quasi-fiscal activities, Table 2b). This suggests that the authorities need to remain vigilant to exposure to contingent fiscal liabilities from quasi-fiscal activities related to state-owned enterprises (SOEs) and the financial sector.

# CONCLUSION

**11. The current DSA, assessed based on thresholds for medium policy performers and the higher discount rate of 5 percent now applied to external debt in LIC DSAs, and also accounting for the lower initial debt position, concludes that Tajikistan's risk of debt distress is reclassified from moderate to low.** All external debt burden indicators are projected to remain below their respective thresholds under both the baseline scenario and stress scenarios. In comparison with the stress scenario in the previous DSA, in which the PV of debt-to-GDP + remittances was projected to breach the indicative threshold for weak policy performers significantly for a protracted period of time, there are no threshold breaches under the stress scenario in the current DSA. However, Tajikistan's external position is projected to remain sensitive to external shocks—particularly interest rate, export, and one time depreciation shocks.

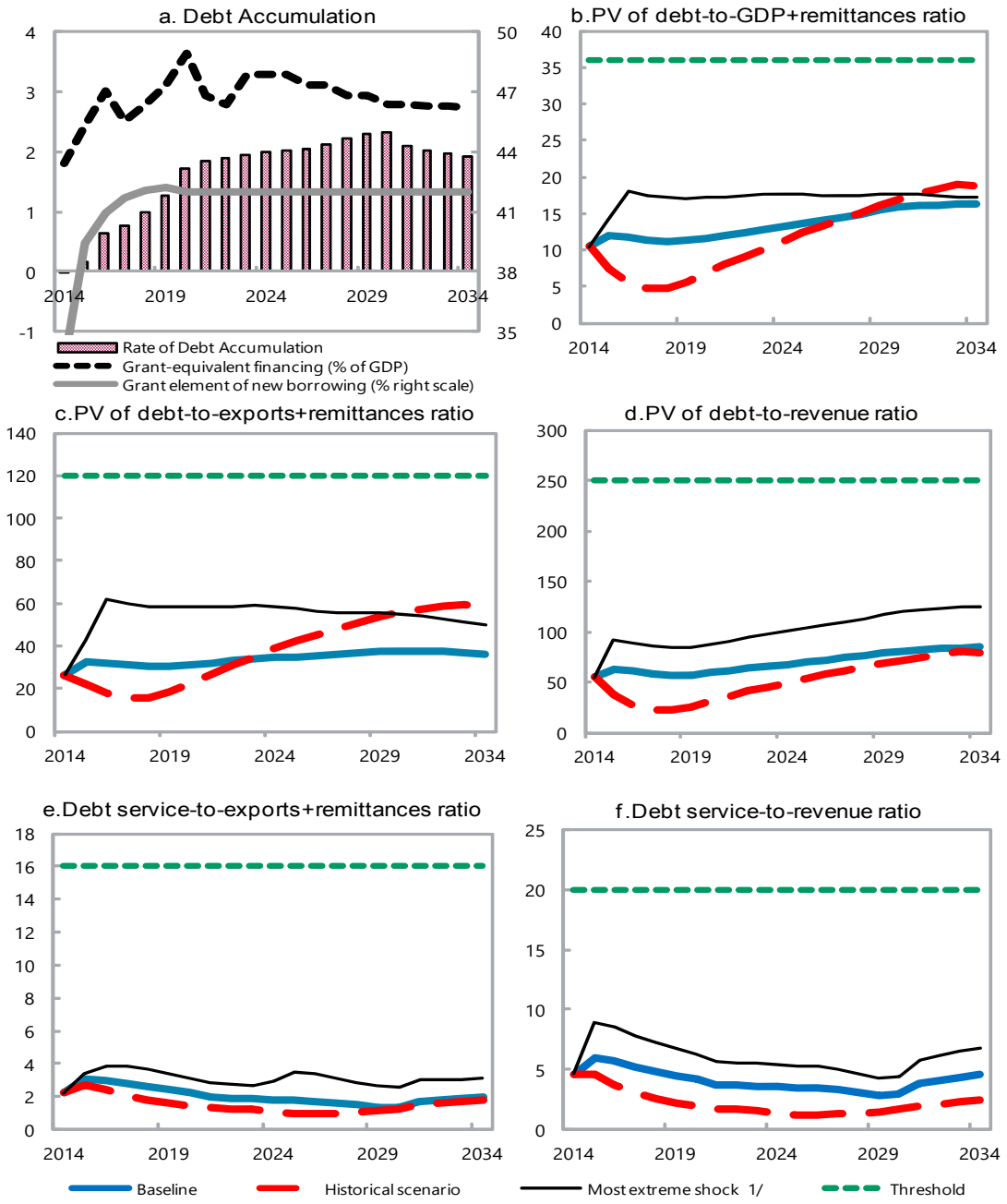
**12. Similar to the previous DSA, standard stress tests indicate that public debt is sensitive to potential shocks relevant for Tajikistan**, namely permanently lower GDP growth (driven by lower growth in Russia) and/or increased debt creating flows (e.g. fiscal costs associated with banks and SOEs).

**13. The current DSA results thus underscore the need for fiscal prudence, caution in contracting and guaranteeing new external debt on non-concessional terms, and minimization of contingent fiscal risks.** Sound macroeconomic policies and acceleration of growth-enhancing structural reforms would also be helpful in maintaining debt sustainability. Going forward, the authorities should halt the practice of directed lending, both to limit contingent fiscal liabilities and to protect the health of banks. Emphasis should also be placed on strengthening debt management capacity by closely monitoring the debts of state-owned enterprises (SOEs) and completing an assessment of, and the plan to address, potential fiscal risks emerging from SOEs and banks (in line with IMF TA recommendations). The authorities

should also develop the domestic debt market as an alternate source of market-based and more transparent financing.

**14. The authorities did not fully concur with the conclusions of the DSA exercise.** They agreed with staff's advice on exercising caution over contracting and guaranteeing new debt on non-concessional terms given existing vulnerabilities and limited buffers. In this vein, the Ministry of Finance acknowledged the need to build greater fiscal buffers and the authorities agreed on the need for greater coordination between the Ministry of Finance and the central bank (NBT) to identify and limit quasi-fiscal activities that ultimately generate unexpected fiscal costs and debt obligations. They have reservations with the DSA shifting them to low risk of debt distress. Upon shifting to medium risk of debt distress in 2013, the authorities now receive around half of their total support from IFIs in the form of grants, with the rest as concessional loans. They are concerned that the further upgrade in their risk rating will reduce their ability to contract external debt on favorable terms from both IFI and bilateral lenders.

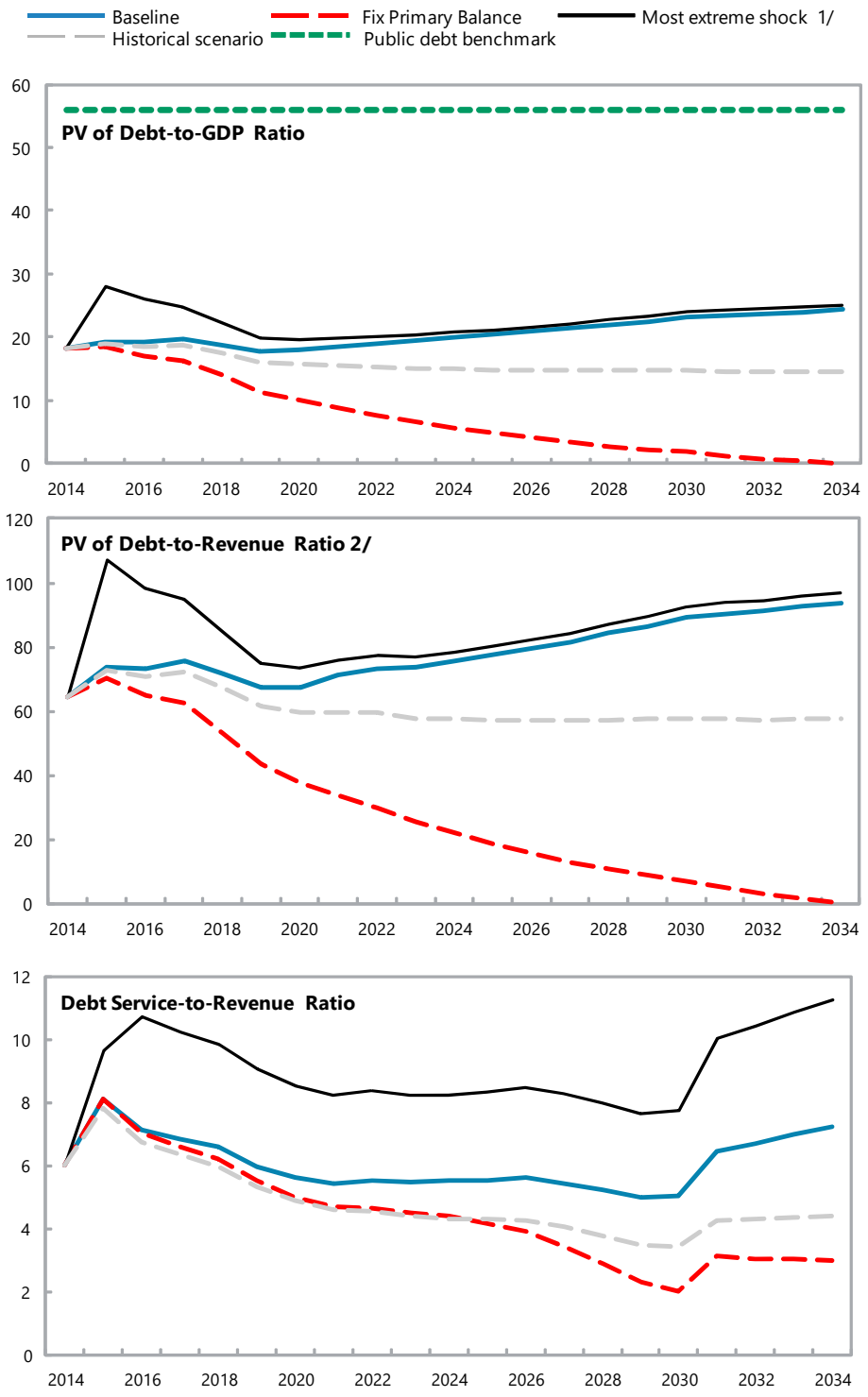
**Figure 1. Tajikistan: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios 2014–2034 1/**




Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Exports shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

**Figure 2. Tajikistan: Indicators of Public Debt Under Alternative Scenarios 2014–2034 1/**




Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

## Table 1a. Tajikistan: External Debt Sustainability Framework, Baseline, Scenario, 2011–2034 1/

(In percent of GDP, unless otherwise indicated)

| | Actual | | | Historical Average 6/ | Standard Deviation 6/ | Projections | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2011 | 2012 | 2013 | | | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2014-2019 Average | 2024 | 2034 | 2020-2034 Average |
| **External debt (nominal) 1/** | **49.9** | **46.6** | **43.6** | | | **42.5** | **46.7** | **47.5** | **47.9** | **48.6** | **49.4** | | **49.4** | **48.6** | |
| *of which: public and publicly guaranteed (PPG)* | 33.1 | 28.4 | 25.3 | | | 24.5 | 25.7 | 26.5 | 27.0 | 27.9 | 28.9 | | 29.7 | 31.3 | |
| Change in external debt | -1.1 | -3.3 | -3.0 | | | -1.0 | 4.2 | 0.8 | 0.4 | 0.7 | 0.9 | | 0.1 | -1.5 | |
| Identified net debt-creating flows | -3.0 | -7.6 | -3.4 | | | 4.2 | 3.0 | 0.8 | -0.9 | -1.9 | -2.7 | | -3.9 | -5.0 | |
| **Non-interest current account deficit** | **3.9** | **1.2** | **1.4** | **1.7** | **5.1** | **7.8** | **5.7** | **4.5** | **3.7** | **3.0** | **2.6** | | **2.2** | **1.3** | 1.8 |
| Deficit in balance of goods and services | 49.3 | 45.9 | 48.8 | | | 46.8 | 32.6 | 29.2 | 27.8 | 26.7 | 25.9 | | 25.9 | 26.7 | |
| Exports | 17.9 | 21.6 | 18.7 | | | 15.4 | 19.1 | 19.5 | 19.8 | 20.5 | 20.8 | | 22.9 | 29.9 | |
| Imports | 67.2 | 67.5 | 67.5 | | | 62.2 | 51.7 | 48.7 | 47.6 | 47.2 | 46.7 | | 48.8 | 56.6 | |
| Net current transfers (negative = inflow) | -45.2 | -44.3 | -46.5 | -37.3 | 11.2 | -38.7 | -26.8 | -24.6 | -24.1 | -23.7 | -23.2 | | -23.1 | -24.6 | -23.5 |
| *of which: official* | -1.8 | -1.2 | -1.5 | | | -0.8 | -1.3 | -1.1 | -0.9 | -0.8 | -0.7 | | -0.4 | -0.2 | |
| Other current account flows (negative = net inflow) | -0.2 | -0.4 | -0.9 | | | -0.2 | -0.1 | -0.1 | 0.0 | -0.1 | -0.2 | | -0.6 | -0.9 | |
| **Net FDI (negative = inflow)** | **-1.0** | **-3.1** | **-1.2** | **-3.2** | **3.7** | **-2.1** | **-2.8** | **-3.2** | **-3.6** | **-3.8** | **-4.2** | | **-5.0** | **-4.9** | -5.0 |
| **Endogenous debt dynamics 2/** | **-6.0** | **-5.7** | **-3.5** | | | **-1.4** | **0.0** | **-0.5** | **-1.0** | **-1.1** | **-1.1** | | **-1.0** | **-1.3** | |
| Contribution from nominal interest rate | 0.9 | 1.3 | 1.5 | | | 1.2 | 1.4 | 1.3 | 1.2 | 1.1 | 1.1 | | 1.2 | 1.0 | |
| Contribution from real GDP growth | -3.3 | -3.2 | -3.1 | | | -2.7 | -1.4 | -1.8 | -2.2 | -2.2 | -2.2 | | -2.3 | -2.3 | |
| Contribution from price and exchange rate changes | -3.6 | -3.8 | -1.9 | | | ... | ... | ... | ... | ... | ... | | ... | ... | |
| **Residual (3-4) 3/** | **1.9** | **4.3** | **0.4** | | | **-5.3** | **1.2** | **-0.1** | **1.3** | **2.6** | **3.6** | | **4.0** | **3.5** | |
| *of which: exceptional financing* | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| PV of external debt 4/ | ... | ... | 33.2 | | | 32.7 | 36.4 | 35.8 | 35.1 | 34.8 | 34.6 | | 36.3 | 38.0 | |
| In percent of exports | ... | ... | 177.3 | | | 212.4 | 190.4 | 183.1 | 177.5 | 169.7 | 166.5 | | 158.5 | 127.3 | |
| **PV of PPG external debt** | **...** | **...** | **14.9** | | | **14.7** | **15.4** | **14.7** | **14.3** | **14.0** | **14.1** | | **16.7** | **20.8** | |
| **In percent of exports** | **...** | **...** | **79.6** | | | **95.4** | **80.5** | **75.5** | **72.2** | **68.6** | **67.8** | | **72.7** | **69.5** | |
| **In percent of government revenues** | **...** | **...** | **60.4** | | | **54.7** | **63.1** | **60.6** | **58.1** | **57.2** | **57.4** | | **67.8** | **84.5** | |
| **Debt service-to-exports ratio (in percent)** | **5.1** | **10.5** | **14.2** | | | **15.1** | **13.6** | **13.0** | **11.9** | **10.9** | **10.3** | | **7.9** | **5.8** | |
| **PPG debt service-to-exports ratio (in percent)** | **5.5** | **6.3** | **8.0** | | | **8.1** | **7.6** | **7.1** | **6.4** | **5.8** | **5.3** | | **3.8** | **3.7** | |
| **PPG debt service-to-revenue ratio (in percent)** | **4.3** | **5.7** | **6.1** | | | **4.7** | **6.0** | **5.7** | **5.2** | **4.9** | **4.5** | | **3.6** | **4.6** | |
| Total gross financing need (Billions of U.S. dollars) | 0.3 | 0.1 | 0.3 | | | 0.8 | 0.5 | 0.4 | 0.2 | 0.2 | 0.1 | | -0.2 | -0.8 | |
| Non-interest current account deficit that stabilizes debt ratio | 5.0 | 4.5 | 4.4 | | | 8.9 | 1.5 | 3.7 | 3.3 | 2.3 | 1.7 | | 2.1 | 2.7 | |
| **Key macroeconomic assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 7.4 | 7.5 | 7.4 | 7.3 | 1.6 | 6.7 | 3.0 | 4.1 | 5.0 | 5.0 | 5.0 | 4.8 | 5.0 | 5.0 | 5.0 |
| GDP deflator in US dollar terms (change in percent) | 7.6 | 8.3 | 4.3 | 10.9 | 10.4 | 1.8 | -10.4 | 3.3 | 3.5 | 3.2 | 3.6 | 0.8 | 3.7 | 3.7 | 3.7 |
| Effective interest rate (percent) 5/ | 2.1 | 3.0 | 3.5 | 4.8 | 5.9 | 3.1 | 3.1 | 3.1 | 2.7 | 2.5 | 2.5 | 2.8 | 2.7 | 2.1 | 2.4 |
| Growth of exports of G&S (US dollar terms, in percent) | 34.5 | 41.0 | -3.1 | 8.4 | 25.6 | -10.6 | 14.5 | 9.9 | 10.0 | 12.2 | 10.5 | 7.8 | 11.4 | 12.1 | 11.6 |
| Growth of imports of G&S (US dollar terms, in percent) | 47.6 | 16.9 | 12.0 | 20.6 | 27.2 | 0.1 | -23.2 | 1.2 | 6.1 | 7.5 | 7.6 | -0.1 | 10.4 | 10.6 | 10.3 |
| Grant element of new public sector borrowing (in percent) | ... | ... | ... | ... | ... | 33.5 | 39.5 | 40.9 | 41.7 | 42.1 | 42.3 | 40.0 | 42.0 | 42.0 | 42.0 |
| Government revenues (excluding grants, in percent of GDP) | 22.6 | 23.6 | 24.6 | | | 26.9 | 24.4 | 24.3 | 24.6 | 24.5 | 24.6 | | 24.6 | 24.6 | 24.6 |
| Aid flows (in Billions of US dollars) 7/ | 0.1 | 0.1 | 0.2 | | | 0.2 | 0.2 | 0.3 | 0.3 | 0.4 | 0.5 | | 0.7 | 1.5 | |
| *of which: Grants* | 0.1 | 0.1 | 0.2 | | | 0.1 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | | 0.3 | 0.6 | |
| *of which: Concessional loans* | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.1 | 0.1 | 0.1 | 0.2 | 0.2 | | 0.4 | 1.0 | |
| Grant-equivalent financing (in percent of GDP) 8/ | ... | ... | ... | | | 1.8 | 2.4 | 3.0 | 2.5 | 2.8 | 3.1 | | 3.3 | 2.7 | 3.0 |
| Grant-equivalent financing (in percent of external financing) 8/ | ... | ... | ... | | | 76.9 | 73.1 | 70.4 | 65.8 | 65.6 | 65.4 | | 62.6 | 58.5 | 60.4 |
| *Memorandum items:* | | | | | | | | | | | | | | | |
| Nominal GDP (Billions of US dollars) | 6.5 | 7.6 | 8.5 | | | 9.2 | 8.5 | 9.2 | 10.0 | 10.8 | 11.7 | | 18.0 | 42.2 | |
| Nominal dollar GDP growth | 15.6 | 16.4 | 12.0 | | | 8.7 | -7.7 | 7.5 | 8.7 | 8.3 | 8.7 | 5.7 | 8.9 | 8.9 | 8.9 |
| PV of PPG external debt (in Billions of US dollars) | | | 1.3 | | | 1.3 | 1.3 | 1.3 | 1.4 | 1.5 | 1.6 | | 3.0 | 8.7 | |
| (PVt-PVt-1)/GDPt-1 (in percent) | | | | | | 0.0 | 0.2 | 0.6 | 0.8 | 1.0 | 1.3 | 0.6 | 2.0 | 1.9 | 2.0 |
| Gross workers' remittances (Billions of US dollars) | 2.8 | 3.6 | 4.1 | | | 3.7 | 2.4 | 2.4 | 2.6 | 2.8 | 3.0 | | 4.6 | 11.6 | |
| PV of PPG external debt (in percent of GDP + remittances) | ... | ... | 10.1 | | | 10.5 | 12.0 | 11.6 | 11.3 | 11.2 | 11.2 | | 13.3 | 16.3 | |
| PV of PPG external debt (in percent of exports + remittances) | ... | ... | 22.3 | | | 26.3 | 32.3 | 32.0 | 31.1 | 30.4 | 30.5 | | 34.3 | 36.3 | |
| Debt service of PPG external debt (in percent of exports + remittances) | ... | ... | 2.2 | | | 2.2 | 3.1 | 3.0 | 2.8 | 2.6 | 2.4 | | 1.8 | 2.0 | |

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and publically guaranteed and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes and other flows.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

## Table 1b. Tajikistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–2034

(In percent)

| | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | **2024** | 2034 |
| **PV of debt-to GDP ratio** | | | | | | | | |
| **Baseline** | 15 | 15 | 15 | 14 | 14 | 14 | **17** | 21 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 15 | 9 | 7 | 6 | 5 | 6 | **12** | 19 |
| A2. New public sector loans on less favorable terms in 2014-2034 2/ | 15 | 15 | 15 | 15 | 16 | 16 | **22** | 30 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 15 | 15 | 14 | 13 | 13 | 13 | **16** | 20 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 15 | 18 | 23 | 22 | 22 | 21 | **22** | 22 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 15 | 13 | 13 | 13 | 13 | 13 | **15** | 19 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 15 | 17 | 18 | 17 | 17 | 17 | **19** | 21 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 15 | 9 | 4 | 4 | 4 | 4 | **8** | 15 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 15 | 22 | 22 | 21 | 21 | 21 | **25** | 31 |
| **PV of debt-to-exports ratio** | | | | | | | | |
| **Baseline** | 95 | 80 | 75 | 72 | 69 | 68 | **73** | 70 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 95 | 48 | 34 | 28 | 27 | 30 | **53** | 65 |
| A2. New public sector loans on less favorable terms in 2014-2034 2/ | 95 | 80 | 79 | 78 | 77 | 79 | **96** | 102 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 95 | 78 | 74 | 71 | 68 | 67 | **72** | 69 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 95 | 131 | 215 | 205 | 194 | 189 | **178** | 134 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 95 | 78 | 74 | 71 | 68 | 67 | **72** | 69 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 95 | 91 | 93 | 89 | 84 | 82 | **82** | 70 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 95 | 65 | 33 | 32 | 32 | 35 | **58** | 87 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 95 | 78 | 74 | 71 | 68 | 67 | **72** | 69 |
| **PV of debt-to-revenue ratio** | | | | | | | | |
| **Baseline** | 55 | 63 | 61 | 58 | 57 | 57 | **68** | 84 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 55 | 37 | 27 | 23 | 22 | 25 | **50** | 79 |
| A2. New public sector loans on less favorable terms in 2014-2034 2/ | 55 | 63 | 63 | 63 | 64 | 67 | **90** | 123 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 55 | 60 | 57 | 55 | 54 | 54 | **64** | 80 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 55 | 74 | 94 | 90 | 88 | 87 | **91** | 89 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 55 | 55 | 54 | 52 | 52 | 52 | **61** | 76 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 55 | 71 | 74 | 71 | 70 | 70 | **77** | 86 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 55 | 35 | 15 | 15 | 15 | 17 | **31** | 60 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 55 | 92 | 89 | 86 | 85 | 85 | **100** | 125 |

## Table 1b. Tajikistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–2034 (continued)

(In percent)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Debt service-to-exports ratio** | | | | | | | | |
| **Baseline** | 8 | 8 | 7 | 6 | 6 | 5 | **4** | 4 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 8 | 6 | 5 | 4 | 3 | 3 | **1** | 2 |
| A2. New public sector loans on less favorable terms in 2014-2034 2/ | 8 | 8 | 7 | 7 | 6 | 6 | **5** | 6 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 8 | 8 | 7 | 6 | 6 | 5 | **4** | 4 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 8 | 11 | 14 | 13 | 12 | 11 | **9** | 8 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 8 | 8 | 7 | 6 | 6 | 5 | **4** | 4 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 8 | 8 | 7 | 7 | 6 | 6 | **5** | 4 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 8 | 9 | 9 | 8 | 7 | 6 | **3** | 4 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 8 | 8 | 7 | 6 | 6 | 5 | **4** | 4 |
| **Debt service-to-revenue ratio** | | | | | | | | |
| **Baseline** | 5 | 6 | 6 | 5 | 5 | 5 | **4** | 5 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 5 | 5 | 4 | 3 | 3 | 2 | **1** | 2 |
| A2. New public sector loans on less favorable terms in 2014-2034 2/ | 5 | 6 | 6 | 5 | 5 | 5 | **5** | 8 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 5 | 6 | 5 | 5 | 5 | 4 | **3** | 4 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 5 | 6 | 6 | 6 | 5 | 5 | **5** | 6 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 5 | 5 | 5 | 5 | 4 | 4 | **3** | 4 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 5 | 6 | 6 | 5 | 5 | 5 | **4** | 5 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 5 | 5 | 4 | 3 | 3 | 3 | **2** | 3 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 5 | 9 | 9 | 8 | 7 | 7 | **5** | 7 |
| *Memorandum item:* | | | | | | | | |
| Grant element assumed on residual financing (i.e., financing required above baseline) 6/ | 40 | 40 | 40 | 40 | 40 | 40 | **40** | 40 |

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

## Table 2a. Tajikistan: Public Sector Debt Sustainability Framework, Baseline, Scenario, 2011–2034

(In percent of GDP, unless otherwise indicated)

| | Actual | | | | | Estimate | Projections | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2011 | 2012 | 2013 | Average 5/ | Standard Deviation 5/ | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2014-19 Average | 2024 | 2034 | 2020-34 Average |
| **Public sector debt 1/** | 35.4 | 32.4 | 29.2 | | | 28.2 | 29.1 | 29.7 | 30.9 | 30.9 | 31.7 | | 32.4 | 34.7 | |
| *of which: foreign-currency denominated* | 33.1 | 28.4 | 25.3 | | | 24.5 | 25.7 | 26.5 | 27.0 | 27.9 | 28.9 | | 29.7 | 31.3 | |
| | 2.3 | 4.0 | 3.9 | | | | | | | | | | 0.0 | 0.0 | |
| Change in public sector debt | -0.9 | -3.1 | -3.2 | | | -1.0 | 0.9 | 0.6 | 1.2 | 0.0 | 0.8 | | 0.3 | 0.3 | |
| Identified debt-creating flows | 1.0 | -5.9 | -2.4 | | | -0.6 | 2.3 | 0.0 | -0.1 | 0.1 | 0.3 | | 0.9 | 0.6 | |
| Primary deficit | 2.4 | -1.3 | -0.4 | 1.7 | 2.6 | -0.8 | 0.9 | 1.6 | 1.7 | 1.9 | 2.2 | 1.2 | 2.7 | 2.6 | 2.7 |
| Revenue and grants | 24.9 | 25.1 | 26.9 | | | 28.4 | 26.2 | 26.5 | 26.1 | 26.3 | 26.5 | | 26.4 | 25.9 | |
| *of which: grants* | 2.3 | 1.5 | 2.3 | | | 1.5 | 1.9 | 2.1 | 1.6 | 1.7 | 1.9 | | 1.9 | 1.3 | |
| Primary (noninterest) expenditure | 27.2 | 23.8 | 26.5 | | | 27.6 | 27.1 | 28.0 | 27.8 | 28.2 | 28.7 | | 29.2 | 28.5 | |
| Automatic debt dynamics | -3.2 | -5.0 | -2.2 | | | 0.0 | 1.3 | -1.8 | -1.9 | -1.9 | -2.0 | | -1.9 | -2.0 | |
| Contribution from interest rate/growth differential | -2.7 | -2.6 | -1.8 | | | -1.7 | -0.8 | -1.2 | -1.5 | -1.5 | -1.6 | | -1.4 | -1.5 | |
| *of which: contribution from average real interest rate* | -0.2 | -0.1 | 0.4 | | | 0.1 | 0.1 | 0.0 | 0.0 | -0.1 | -0.1 | | 0.1 | 0.2 | |
| *of which: contribution from real GDP growth* | -2.5 | -2.5 | -2.2 | | | -1.8 | -0.8 | -1.1 | -1.4 | -1.5 | -1.5 | | -1.5 | -1.6 | |
| Contribution from real exchange rate depreciation | -0.5 | -2.4 | -0.4 | | | 1.7 | 2.0 | -0.6 | -0.4 | -0.4 | -0.4 | | ... | ... | |
| Other identified debt-creating flows | 1.9 | 0.4 | 0.2 | | | 0.2 | 0.2 | 0.1 | 0.1 | 0.1 | 0.1 | | 0.0 | 0.0 | |
| Privatization receipts (negative) | 1.9 | 0.4 | 0.2 | | | 0.2 | 0.2 | 0.1 | 0.1 | 0.1 | 0.1 | | 0.0 | 0.0 | |
| Recognition of implicit or contingent liabilities | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Debt relief (HIPC and other) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Other (specify, e.g. bank recapitalization) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Residual, including asset changes | -1.9 | 2.9 | -0.8 | | | -0.4 | -1.4 | 0.6 | 1.3 | -0.1 | 0.5 | | -0.6 | -0.3 | |
| **Other Sustainability Indicators** | | | | | | | | | | | | | | | |
| **PV of public sector debt** | ... | ... | 18.8 | | | 18.4 | 18.8 | 17.9 | 18.1 | 17.0 | 16.8 | | 19.4 | 24.1 | |
| *of which: foreign-currency denominated* | ... | ... | 14.9 | | | 14.7 | 15.4 | 14.7 | 14.3 | 14.0 | 14.1 | | 16.7 | 20.8 | |
| *of which: external* | ... | ... | 14.9 | | | 14.7 | 15.4 | 14.7 | 14.3 | 14.0 | 14.1 | | 16.7 | 20.8 | |
| PV of contingent liabilities (not included in public sector debt) | ... | ... | ... | | | ... | ... | ... | ... | ... | ... | | ... | ... | |
| Gross financing need 2/ | 3.9 | 0.6 | 2.0 | | | 0.9 | 3.0 | 3.4 | 3.4 | 3.7 | 3.8 | | 4.2 | 4.7 | |
| PV of public sector debt-to-revenue and grants ratio (in percent) | ... | ... | 69.6 | | | 64.6 | 71.5 | 67.8 | 69.2 | 64.8 | 63.6 | | 73.3 | 93.1 | |
| PV of public sector debt-to-revenue ratio (in percent) | ... | ... | 76.2 | | | 68.3 | 77.0 | 73.8 | 73.6 | 69.4 | 68.6 | | 78.9 | 98.1 | |
| *of which: external 3/* | ... | ... | 60.4 | | | 54.7 | 63.1 | 60.6 | 58.1 | 57.2 | 57.4 | | 67.8 | 84.5 | |
| Debt service-to-revenue and grants ratio (in percent) 4/ | 6.0 | 7.7 | 8.9 | | | 6.0 | 8.1 | 7.1 | 6.8 | 6.6 | 6.0 | | 5.5 | 7.2 | |
| Debt service-to-revenue ratio (in percent) 4/ | 6.6 | 8.2 | 9.7 | | | 6.4 | 8.7 | 7.8 | 7.3 | 7.0 | 6.4 | | 5.9 | 7.6 | |
| Primary deficit that stabilizes the debt-to-GDP ratio | 3.3 | 1.7 | 2.8 | | | 0.2 | 0.0 | 1.0 | 0.5 | 1.9 | 1.4 | | 2.5 | 2.3 | |
| **Key macroeconomic and fiscal assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 7.4 | 7.5 | 7.4 | 7.3 | 1.6 | 6.7 | 3.0 | 4.1 | 5.0 | 5.0 | 5.0 | 4.8 | 5.0 | 5.0 | 5.0 |
| Average nominal interest rate on forex debt (in percent) | 3.0 | 2.9 | 4.4 | 2.7 | 0.8 | 2.3 | 2.3 | 2.0 | 1.6 | 1.4 | 1.4 | 1.8 | 2.1 | 2.4 | 2.0 |
| Average real interest rate on domestic debt (in percent) | -7.3 | -1.3 | -1.0 | -5.5 | 6.4 | -1.5 | -1.0 | -0.9 | 1.0 | 1.5 | 1.5 | 0.1 | 1.5 | 0.9 | 1.5 |
| Real exchange rate depreciation (in percent, + indicates depreciation) | -1.6 | -7.7 | -1.5 | -6.6 | 10.0 | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| Inflation rate (GDP deflator, in percent) | 13.3 | 11.9 | 4.3 | 15.7 | 7.6 | 5.5 | 12.7 | 8.0 | 6.9 | 6.0 | 6.0 | 7.5 | 6.0 | 6.0 | 6.0 |
| Growth of real primary spending (deflated by GDP deflator, in percent) | 16.8 | -6.0 | 19.5 | 3.1 | 8.2 | 11.1 | 1.1 | 7.7 | 4.1 | 6.6 | 6.7 | 6.2 | 5.0 | 4.9 | 5.0 |
| Grant element of new external borrowing (in percent) | ... | ... | ... | ... | ... | 33.5 | 39.5 | 40.9 | 41.7 | 42.1 | 42.3 | 40.0 | 42.0 | 42.0 | ... |

Sources: Country authorities; and staff estimates and projections.

1/ Public sector is defined as nonfinancial public sector. Gross debt is used.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

## Table 2b. Tajikistan: Sensitivity Analysis for Key Indicators of Public Debt 2014–2034

| | Projections | | | | | | |
|---|---|---|---|---|---|---|---|
| | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2034 |
| **PV of Debt-to-GDP Ratio** | | | | | | | |
| **Baseline** | 18 | 19 | 18 | 18 | 17 | 17 | 24 |
| **A. Alternative scenarios** | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 18 | 18 | 17 | 17 | 16 | 15 | 14 |
| A2. Primary balance is unchanged from 2014 | 18 | 18 | 16 | 14 | 12 | 10 | -1 |
| A3. Permanently lower GDP growth 1/ | 18 | 19 | 18 | 19 | 18 | 18 | 34 |
| **B. Bound tests** | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016 | 18 | 18 | 16 | 16 | 14 | 14 | 16 |
| B2. Primary balance is at historical average minus one standard deviations in 2015-2016 | 18 | 21 | 21 | 21 | 20 | 20 | 25 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 18 | 19 | 19 | 18 | 16 | 15 | 14 |
| B4. One-time 30 percent real depreciation in 2015 | 18 | 28 | 25 | 23 | 20 | 19 | 25 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2015 | 18 | 24 | 23 | 23 | 22 | 22 | 26 |
| **PV of Debt-to-Revenue Ratio 2/** | | | | | | | |
| **Baseline** | 65 | 72 | 68 | 69 | 65 | 64 | 93 |
| **A. Alternative scenarios** | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 65 | 71 | 66 | 66 | 60 | 58 | 55 |
| A2. Primary balance is unchanged from 2014 | 65 | 68 | 59 | 55 | 46 | 39 | -4 |
| A3. Permanently lower GDP growth 1/ | 65 | 72 | 69 | 71 | 67 | 67 | 131 |
| **B. Bound tests** | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016 | 65 | 69 | 62 | 61 | 55 | 52 | 62 |
| B2. Primary balance is at historical average minus one standard deviations in 2015-2016 | 65 | 79 | 81 | 82 | 77 | 75 | 98 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 65 | 74 | 72 | 70 | 63 | 59 | 55 |
| B4. One-time 30 percent real depreciation in 2015 | 65 | 105 | 93 | 88 | 78 | 71 | 96 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2015 | 65 | 92 | 88 | 89 | 84 | 82 | 101 |
| **Debt Service-to-Revenue Ratio 2/** | | | | | | | |
| **Baseline** | 6 | 8 | 7 | 7 | 7 | 6 | 7 |
| **A. Alternative scenarios** | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 6 | 8 | 7 | 6 | 6 | 5 | 4 |
| A2. Primary balance is unchanged from 2014 | 6 | 8 | 7 | 7 | 6 | 5 | 3 |
| A3. Permanently lower GDP growth 1/ | 6 | 8 | 7 | 7 | 7 | 6 | 9 |
| **B. Bound tests** | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016 | 6 | 8 | 7 | 6 | 6 | 6 | 6 |
| B2. Primary balance is at historical average minus one standard deviations in 2015-2016 | 6 | 8 | 7 | 7 | 7 | 6 | 8 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 6 | 8 | 7 | 7 | 6 | 6 | 6 |
| B4. One-time 30 percent real depreciation in 2015 | 6 | 10 | 11 | 10 | 10 | 9 | 11 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2015 | 6 | 8 | 8 | 8 | 7 | 6 | 8 |

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.